

Apoptosis as Controlled Syneresis of a Confined Hydrogel System

Mechanism, Stability Conditions, and Controlled Transition

ABSTRACT

This paper proposes a physical reinterpretation of apoptosis as controlled syneresis in a confined hydrogel system. The cell is modelled as a polymer network saturated with water and enclosed by an elastic membrane. Apoptosis is not merely a biochemical program, but a directed transition into a stable condensed state, characterised by volume reduction, loss of free water, and preservation of network connectivity and membrane integrity.

The key morphological features of apoptosis — in particular apoptotic volume decrease — correspond directly to the mechanics of syneresis in a confined gel [1–3]. A minimal free energy model is introduced incorporating osmotic pressure, elastic energy, membrane tension, network connectivity, and available energy.

1. INTRODUCTION

Apoptosis is traditionally described in biochemical terms: caspase activation, cytoskeletal remodelling, and regulated degradation of cellular components [9]. Yet one of its earliest and most universal features is physical — a pronounced decrease in cell volume, accompanied by water loss and cytoplasmic condensation [1–4].

From a physical perspective, the cell is a confined hydrogel: a polymer network saturated with water and enclosed by an elastic membrane. In polymer physics, *syneresis* denotes the contraction of a gel with expulsion of solvent [5–7].

Is apoptosis primarily a biochemical program that happens to shrink the cell, or is it a physically constrained transition that biochemistry merely orchestrates?

In this work, we propose that apoptosis occupies a well-defined region of the syneresis-like collapse trajectory in the state space of a confined active hydrogel system.

2. STATE VARIABLES

The state of the cell is described by: degree of hydration W ; volume V ; osmotic pressure; elastic energy of the intracellular network; membrane tension; network connectivity C ; available energy E ; free energy F .

The free energy is decomposed as:

$$F = F_{\text{osmotic}} + F_{\text{elastic}} + F_{\text{membrane}}$$

Connectivity C is treated as a dynamical variable, moving the model beyond passive gels into the regime of active matter.

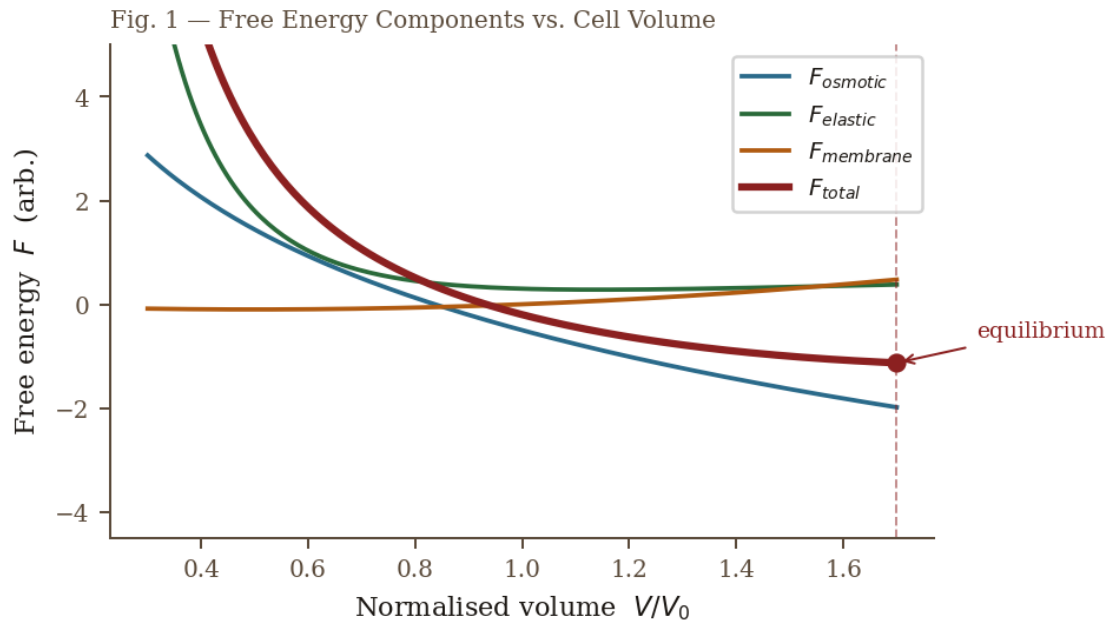


Fig. 1 — Contributions of osmotic, elastic, and membrane terms to total free energy F as a function of normalised cell volume V/V_0 . The minimum of F defines the stable equilibrium state.

3. STABILITY CONDITIONS

The viable state corresponds to a local minimum of free energy. At equilibrium, osmotic pressure is balanced by elastic and membrane contributions:

$$P_{\text{osmotic}} = P_{\text{elastic}}(C) + P_{\text{membrane}}(\sigma)$$

This condition defines a stability domain in state space. Its boundaries are probabilistic, reflecting fluctuations in ion flux, metabolism, and cytoskeletal dynamics [1–4].

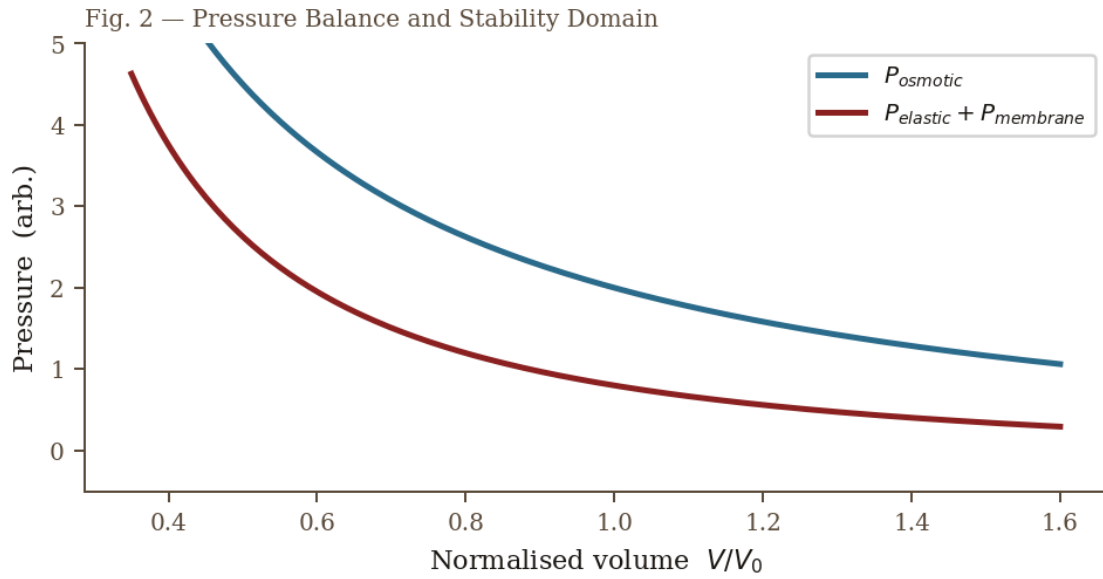


Fig. 2 — Osmotic and elastic+membrane pressure vs. normalised volume. Intersection defines equilibrium; the shaded band marks the stability domain.

4. SYNERESIS AS A PHYSICAL TRANSITION

Syneresis is driven by the dominance of elastic forces over osmotic expansion [5,6]:

$$P_{elastic}(C) > P_{osmotic}(W)$$

In a confined system, this does not lead to collapse, but to a new stable equilibrium at smaller volume, following a specific direction: decreasing hydration and increasing effective network constraint.

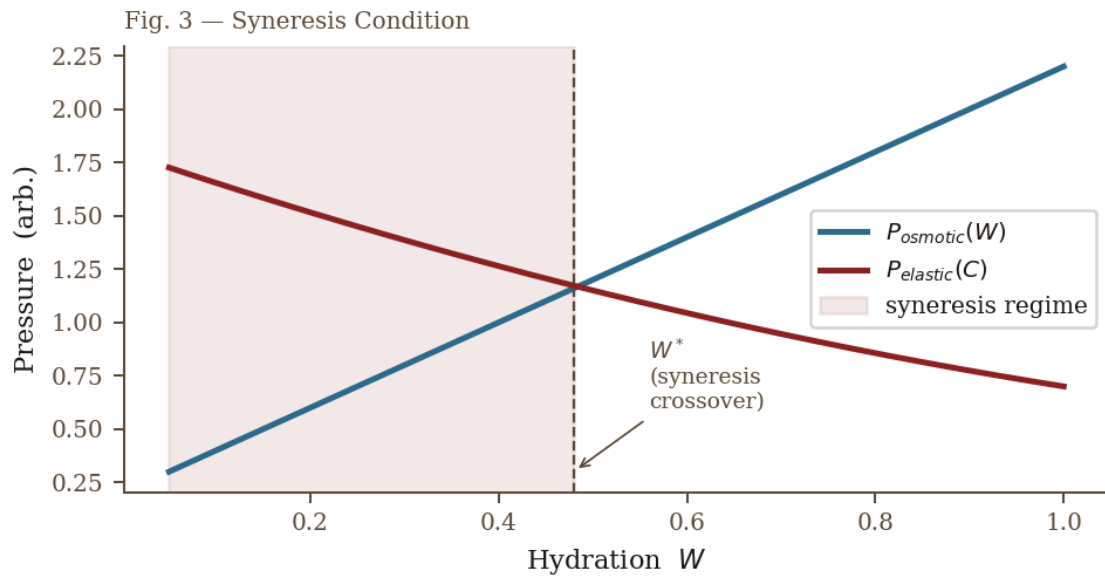


Fig. 3 — Elastic vs. osmotic pressure as a function of hydration W . Above the crossover W^* , elastic pressure exceeds osmotic — the syneresis regime. Apoptotic volume decrease (AVD) drives W into this regime in a controlled manner.

5. APOPTOSIS AS CONTROLLED SYNERESIS

Apoptosis is a regulated traversal of the syneresis-like trajectory:

$$(W_{\blacksquare}, V_{\blacksquare}, C_{\blacksquare}, E_{\blacksquare}) \rightarrow (W_a, V_a, C_a, E_a) \text{ with } W_a < W_{\blacksquare}, V_a < V_{\blacksquare}, d^2F/dV^2 > 0$$

The cell loses water and volume, but remains within the stability domain. This is not a passive collapse, but a guided descent along a physically constrained pathway.

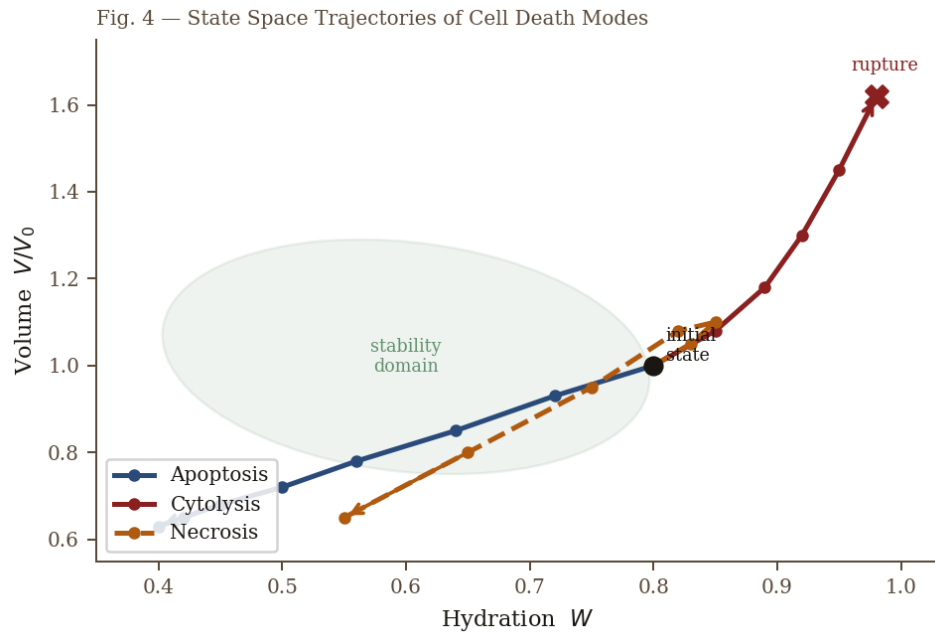


Fig. 4 — Trajectories in the hydration–volume plane. Apoptosis (blue): controlled descent within the stability domain. Cytolysis (red): expansion toward membrane rupture. Necrosis (orange dashed): rapid exit from the stability domain.

5.1 Biochemistry as Trajectory Control

Biochemical processes act as regulators of the path through state space: ion channels reduce osmotic pressure [1–4]; cytoskeletal remodelling alters elastic resistance [8]; membrane changes modify tension.

Biochemistry specifies the trajectory; physics defines the accessible landscape.

5.2 The Role of Energy

Energy does not trigger apoptosis; it maintains the system within a controllable region. Energy depletion may redirect the trajectory into necrotic destabilisation — a physical interpretation of mixed apoptotic–necrotic phenotypes.

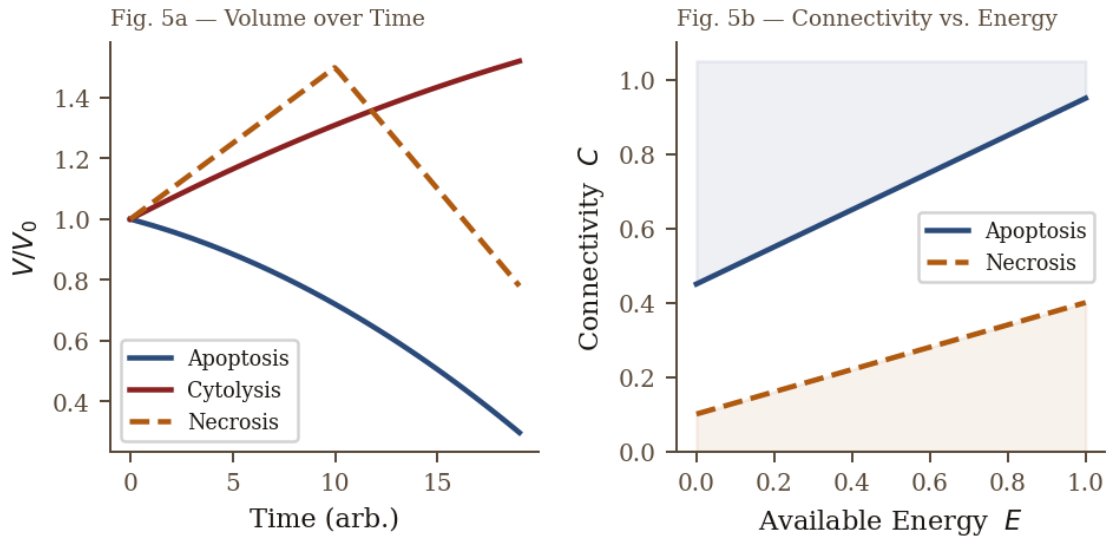


Fig. 5a (left) — Normalised cell volume V/V_0 vs. time for three death modes. Fig. 5b (right) — Network connectivity C as a function of available energy E . Apoptosis maintains connectivity; necrosis loses it rapidly.

6. SEPARATION OF CELL DEATH REGIMES

The state space admits distinct exit trajectories — apoptosis, cytolysis, and necrosis — which are topologically distinct modes of failure, not variations of the same process.

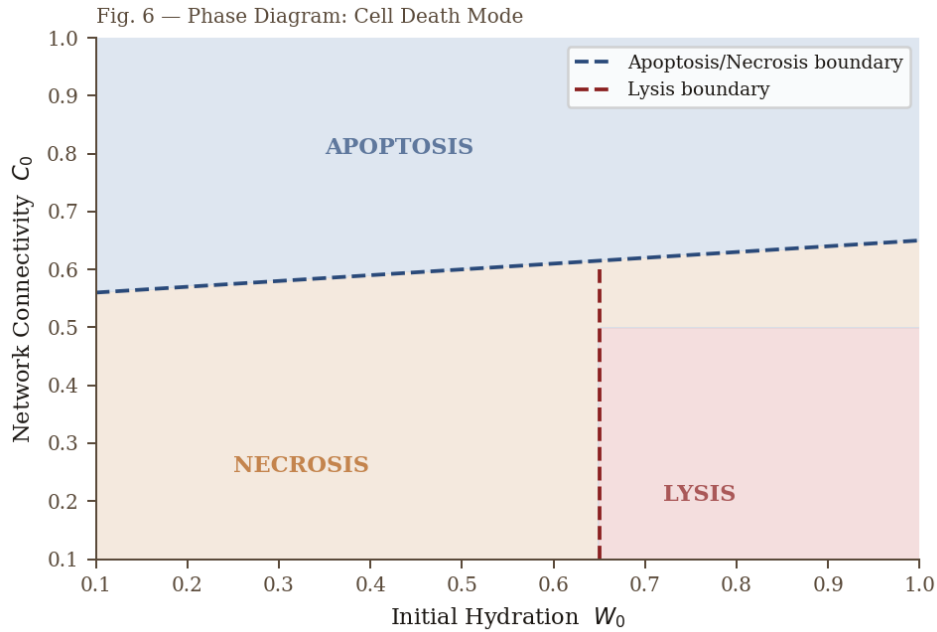


Fig. 6 — Phase diagram: cell death mode as a function of initial connectivity C_0 and hydration W_0 . Cells with higher hydration and weaker connectivity are closer to the lytic boundary; stronger networks bias toward controlled collapse.

7. BIOLOGICAL IMPLICATIONS

The mode of cell death is determined not only by molecular signals, but by the position of the cell in physical state space. This may explain why some cellular states resist apoptosis while others undergo it readily [9].

8. DISCUSSION

We do not claim that apoptosis is reducible to a passive physical process. Rather, the physical trajectory of apoptosis is constrained in a way equivalent to controlled syneresis in a confined active hydrogel. This shift in perspective unifies different death modes within a single framework, provides experimentally testable predictions, and introduces a natural language for describing stability, reversibility, and failure.

9. CONCLUSION

Apoptosis can be understood as a controlled transition within the stability domain of a confined hydrogel system — a specific region of the syneresis-like trajectory, distinct from lytic and necrotic failure modes.

This interpretation does not replace biochemistry. It constrains it.

References

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